Change-Point Detection in Hierarchical Circadian Models

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Abstract

This paper addresses the problem of change-point detection on sequences of high-dimensional and heterogeneous observations, which also possess a periodic temporal structure. Due to the dimensionality problem, when the time between change points is of the order of the dimension of the model parameters, drifts in the underlying distribution can be misidentified as changes. To overcome this limitation, we assume that the observations lie in a lower-dimensional manifold that admits a latent variable representation. In particular, we propose a hierarchical model that is computationally feasible, widely applicable to heterogeneous data and robust to missing instances. Additionally, the observations' periodic dependencies are captured by non-stationary periodic covariance functions. The proposed technique is particularly well suited to (and motivated by) the problem of detecting

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changes in human behavior using smartphones and its application to relapse detection in psychiatric patients. Finally, we validate the technique on synthetic examples and we demonstrate its utility in the detection of behavioral changes using real data acquired by smartphones.
Keywords: Change-point detection, circadian models, heterogeneous data, latent variable models, non-stationary periodic covariance functions.

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1. Introduction

Change-point detection (CPD) consists of locating abrupt transitions in the generative model of a sequence of observations. Detecting these abrupt transitions or change points (CPs) has been long studied and appears in a vast amount of real-world scenarios, for instance, investment strategies, the analysis of social networks or cognitive radio in signal processing.

The problem of CPD can be faced from a statistical point of view, which typically requires a model defined a priori, or alternatively, using model-free methods. The statistical approaches can be further divided between frequentist and Bayesian techniques. The main idea behind frequentist methods is to derive a measure, usually based on likelihood ratio tests [21, 19], between the pre-change and post-change distributions and compare it with a threshold. On the other hand, Bayesian approaches are based on assigning a prior distribution over the change points, or a proper surrogate, and derive its posterior [2]. Typically, these last models focus on batch settings with unknown number of changes [12]. In contrast, online methods have been proposed based on particle filters [13] or the Bayesian online change-point detection (BOCPD) model of [1].

In this work we focus on a novel application of CPD, namely, the problem of modeling and detecting changes in human behavior. CPD can be used to detect behavioral changes, which in patients with prevalent chronic disorders (e. g., schizophrenia or chronic depression) may be signals of future relapses [3]. Other works, like [5], study the rapid detection of suicidal behavior based on pose representations of patients. Here, among the alternatives to monitor psychiatric patients' symptoms, we consider smartphone-based monitoring stands [29]. The ubiquitous presence of smartphones is the key point and the large amount of data gathered by these devices opens new opportunities in this area. For instance, smartphone observations range from inertial sensors measurements or GPS information timestamps to duration of calls or app usage log. However, all these monitoring traces pose some challenges that have not been considered in detail within the CPD framework. First, the observations provided by smartphones are high-dimensional and heterogeneous. Second, since humans are strictly conditioned by 24-hour periods, also known as circadian rhythm, samples usually present some underlying periodicities [10]. Finally, since some of these sensors may fail or be turned off, it is common to have missing values in the data.

A large portion of CPD algorithms proposed in the literature do not consider the aforementioned features of data. Concretely, the general assumption of CPD models is to consider low-dimensional and homogenous datasets [1, 13]. However, these models will not work as the detection of behavioral changes using smartphones usually involve observations that are high-dimensional and heterogeneous, i.e., they are composed by a mixture of continuous, categorical, binary or discrete variables. Typically, heterogenous

observations are difficult to deal with [22], since an improper learning process would yield a system that mainly uses only one of the different data types. Moreover, the CPD problem becomes even more difficult when observations possess temporal structure, that is, they deviate from the independent and identically distributed (i. i. d.) assumption.

This case is also studied in this work, where there exists complex temporal structure, which in our setting is induced by the circadian rhythm. The works [25, 4] also consider CPD on non-stationary sequential data, but our model captures periodic patterns, similarly to the approach in [30].

This paper addresses the problem using a Bayesian CPD approach. This sort of methods provide a general measure of uncertainty, which is critical for us, due to the need of reliable decision-making algorithms in the mental-health context. There has been a considerable effort in Bayesian estimation of change points on different domains. For example, [18] introduced an online cumulative sum detector that finds changes on categorical time series. Alternative approaches are based on mixture models [20, 7], hidden Markov models (HMM) [11], nearest neighbor methods [24] or support vector machines (SVM) [27].

We generalize the BOCPD algorithm [1] to address the challenges raised by the application of CPD in psychiatry. Several extensions of the BOCPD algorithm exist as well, including adaptive sequential methods [31] or the Gaussian Process BOCPD model [26]. The proposed generalization of the BOCPD algorithm considers the CPD problem on heterogeneous and high-dimensional sequences through hierarchical models within latent variables. In Section 2, the hierarchical extension of the BOCPD method is described. First,

we propose a general latent variable model and, later on, due to the complexity induced by arbitrary latent variables, we particularize it to categorical ones. This is also well justified by the proposed application since these categorical latent variables represent what we could call *type of day*, i.e., one day could correspond to a typical workday, another one to a weekend, etc. Thus, the proposed method is actually detecting changes in the distributions of such types of days. For instance, a change point should be detected if the category that represents typical workdays disappears. Moreover, for the hierarchical model we also present the inference procedure and a simplified model, as well as an approach to consider missing data. The association between observations and types of days is presented in Section 3, where a clustering technique based on mixtures of Gaussians is described. Interestingly, the covariance matrix of the Gaussian random vectors will be designed to capture the circadian features induced by human behavior. We develop the inference based on the expectation-maximization (EM) algorithm [8], which also allows us to handle missing data. Finally, the performance of the proposed method is evaluated in Section 4 over synthetic experiments and real data acquired by smartphones.

2. Change-Point Detection in Hierarchical Models

This section generalizes the method proposed in [1] to consider hierarchical models, which allow us to deal with high-dimensional observations. The work in [1] considers that the sequence of observations, $\mathbf{X}_{1:t}$, may be divided into non-overlapping partitions separated by change points. With a few exceptions, such as [26], the work in [1] and its generalizations assume that the data within each partition is i. i. d. according to some generative distribution $p(\mathbf{x}_t|\boldsymbol{\theta}_t)$ with

unknown parameters $\boldsymbol{\theta}_t$. Hence, these parameters are constant within each partition but they change between change points, i.e.,

$$\boldsymbol{\theta}_t = \begin{cases} \boldsymbol{\theta}_a, & t < \mathrm{CP}_1, \\ \boldsymbol{\theta}_b, & \mathrm{CP}_1 \leq t \leq \mathrm{CP}_2, \\ & \vdots \end{cases} \tag{1}$$

where CP_i is the time index of the ith change point and $\boldsymbol{\theta}_a \neq \boldsymbol{\theta}_b \neq \cdots$.[1] Moreover, the run length was defined in [1] as an auxiliary variable to denote when change points appear: $r_t = 0$ if there is a CP at time t and $r_t = r_t + 1$ if there is no CP. Hence, the run length r_t counts the number of time steps since the last CP. The objective of the BOCPD method is therefore to perform the recursive computation of $p(r_t|\mathbf{X}_{1:t})$. From this posterior distribution, it is possible to estimate the run length and as a consequence the CPs using, for instance, the maximum-a-posteriori (MAP) criterion. However, due to the recursive nature of the $p(r_t|\mathbf{X}_{1:t})$ computation, the algorithm may be unable to accumulate enough probability mass on low values of r_t when the time between CPs becomes of the order of magnitude (or less) of the number of the model parameters in $\boldsymbol{\theta}_t$. This makes almost impossible to detect CPs in high-dimensional observations.

To overcome this limitation, we further assume that even if the data are high-dimensional, they belong to a low-dimensional manifold as follows

$$p(\mathbf{x}_t|\boldsymbol{\theta}_t) = \int p(\mathbf{x}_t|\mathbf{z}_t)p(\mathbf{z}_t|\boldsymbol{\theta}_t)d\mathbf{z}_t, \tag{2}$$

where $\mathbf{z}_t$ is the vector of latent variables corresponding to $\mathbf{x}_t$. Now, the

[1] Note that not all components of $\boldsymbol{\theta}_t$ need to change between CPs.

parameters that change between CPs parametrize the distribution of the latent variables $p(\mathbf{z}_t|\boldsymbol{\theta}_t)$, and since these variables have a dimensionality smaller than that of $\mathbf{x}_t$, the size of $\boldsymbol{\theta}_t$ is also reduced.

For the time being, we consider that $\mathbf{z}_t$ may be either continuous or discrete, and $p(\mathbf{x}_t|\mathbf{z}_t)$ is fixed and known a priori.[2] For this hierarchical model, we are still interested in the posterior $p(r_t|\mathbf{X}_{1:t})$, and we therefore need to extend the recursive computation of [1] to account for the sequence of latent vectors, $\mathbf{Z}_{1:t}$. The posterior $p(r_t|\mathbf{X}_{1:t})$ can be computed from the joint distribution $p(r_t, \mathbf{Z}_{1:t}, \mathbf{X}_{1:t}, \boldsymbol{\theta}_t)$ as

$$p(r_t|\mathbf{X}_{1:t}) = \frac{p(r_t, \mathbf{X}_{1:t})}{\sum_{r_t} p(r_t, \mathbf{X}_{1:t})}, \tag{3}$$

where

$$p(r_t, \mathbf{X}_{1:t}) = \iint p(r_t, \mathbf{Z}_{1:t}, \mathbf{X}_{1:t}, \boldsymbol{\theta}_t)\ d\boldsymbol{\theta}_t d\mathbf{Z}_{1:t}. \tag{4}$$

This procedure can be quite involved, mainly due to the integrals in (4), and it is not trivial to achieve a recursive expression. To obtain the required recursitivity, we factorize the joint distribution in (4) as

$$p(r_t, \mathbf{Z}_{1:t}, \mathbf{X}_{1:t}, \boldsymbol{\theta}_t) = p(\mathbf{X}_{1:t}|\mathbf{Z}_{1:t})p(r_t, \mathbf{Z}_{1:t}, \boldsymbol{\theta}_t), \tag{5}$$

where we have exploited that $p(\mathbf{x}_t|\mathbf{z}_t)$ is assumed fixed and known. Now, we continue by marginalizing out the parameters $\boldsymbol{\theta}_t$ in (5), which yields

$$p(r_t, \mathbf{Z}_{1:t}, \mathbf{X}_{1:t}) = p(\mathbf{X}_{1:t}|\mathbf{Z}_{1:t})p(r_t, \mathbf{Z}_{1:t}). \tag{6}$$

[2] In Section 3 we relax this restriction and use a conditional distribution parametrized by some *unknown* parameters $\boldsymbol{\phi}_t$.

Note that the recursive nature of the algorithm is a direct consequence of the factorization in (5), leading to

$$p(r_t, \mathbf{Z}_{1:t}, \boldsymbol{\theta}_t) = \sum_{r_{t-1}} p(r_t|r_{t-1}) p(\mathbf{z}_t|\boldsymbol{\theta}_t) p(\boldsymbol{\theta}_t|r_{t-1}, \mathbf{Z}_{1:t-1}^{(r)}) p(r_{t-1}, \mathbf{Z}_{1:t-1}), \tag{7}$$

where $\mathbf{Z}_{1:t-1}^{(r)}$ denotes the subset of latent variables for a given partition [1]. More details about the complete derivation of (7) can be found in Appendix A, included in the supplementary material. Importantly, the last term in the right-hand side of (7) can be obtained as

$$p(r_{t-1}, \mathbf{Z}_{1:t-1}) = \int p(r_{t-1}, \mathbf{Z}_{1:t-1}, \boldsymbol{\theta}_{t-1}) d\boldsymbol{\theta}_{t-1}. \tag{8}$$

The prior of r_t conditioned to r_{t-1}, also known as the change-point prior, is given by

$$p(r_t|r_{t-1}) = \begin{cases} H(r_{t-1}+1), & r_t = 0, \\ 1 - H(r_{t-1}+1), & r_t = r_{t-1}+1, \end{cases} \tag{9}$$

where $H(\tau)$ is the *hazard* function and we take it to be constant with a given time-scale hyperparameter τ. Finally, $p(\boldsymbol{\theta}_t|r_{t-1}, \mathbf{Z}_{1:t-1}^{(r)})$ is the posterior of the parameters at time t given the previous run length and the corresponding partition of the latent variables $\mathbf{Z}_{1:t-1}^{(r)}$.

Combining all results, the probability $p(r_t, \mathbf{Z}_{1:t})$ becomes

$$p(r_t, \mathbf{Z}_{1:t}) = \sum_{r_{t-1}} p(r_t|r_{t-1}) \pi_t^{(r)} p(r_{t-1}, \mathbf{Z}_{1:t-1}), \tag{10}$$

where the predictive posterior [1, 31] is given by

$$\pi_t^{(r)} = p(\mathbf{z}_t|r_{t-1}, \mathbf{Z}_{1:t-1}^{(r)}) = \int p(\mathbf{z}_t|\boldsymbol{\theta}_t) p(\boldsymbol{\theta}_t|r_{t-1}, \mathbf{Z}_{1:t-1}^{(r)}) d\boldsymbol{\theta}_t. \tag{11}$$

So far, we have obtained a recursive expression, but we face the first problem due to the marginalization of $\boldsymbol{\theta}_t$, namely, the exact evaluation of $\pi_t^{(r)}$ is often intractable for non-Gaussian likelihoods and may require numerical solutions. Actually, the computation of $\pi_t^{(r)}$ is a well-known issue for every extension of [1]. The second problem is the marginalization over the sequence of latent variables $\mathbf{Z}_{1:t}$ in (4). To overcome this problem, we propose to use a simple latent class model, where the latent variables are given by $z_t \in \{1, 2, \ldots, K\}$, with K being the number of classes. Moreover, this choice also results in a small number of unknown parameters, leading to better performance of the CPD algorithm. On top of that, a latent class model also simplifies the interpretability of the algorithm, as we will see in Section 4. To sum up, $\mathbf{x}_t$ has now a single discrete representation z_t, which indicates the class it belongs to. Thus, we need to compute $p(r_t, \mathbf{z}_{1:t}, \mathbf{X}_{1:t}, \boldsymbol{\theta}_t)$, where $\mathbf{z}_{1:t} = [z_1, \ldots, z_t]^\top$ is the *univariate* sequence of latent class indicators. With this in mind, we depict in Figure 1 the proposed hierarchical model.

Due to the recursive dependence in (10), marginalizing the whole sequence of latent class variables yields a combinatorial problem, which can be computationally challenging. Concretely, for sequences of length T and with K classes, the evaluation of $p(r_t, \mathbf{X}_{1:t})$ requires $\mathcal{O}(K^T)$ operations. Therefore, it may not be possible to compute it for long observation periods, T, and/or a large number of classes, K.

2.1. Simplified Hierarchical Model

We have shown how the computational complexity of the hierarchical model grows exponentially as T and K increase, which becomes prohibitive very quickly. However, we are still interested in the latent variable representa-

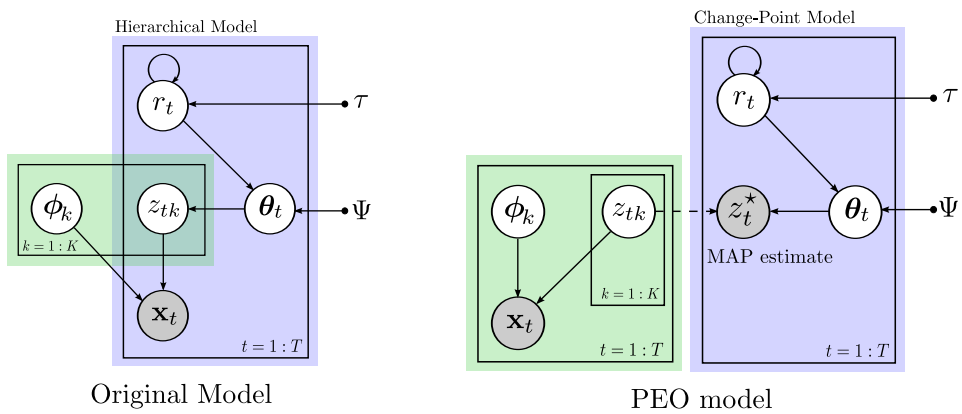

Figure 1: Graphical representation of the original **(left)** and simplified **(right)** hierarchical models. Shaded nodes correspond to observed variables and the dashed line denotes how the posterior probabilities of z_t are observed by the change-point model (blue). The variable, $z_t^\star$ is a point estimate which takes the z_{tk} value with highest probability. Additionally, Ψ denotes the fixed hyperparameters from the corresponding priors placed on $\boldsymbol{\theta}_t$.

tion for the problem of change-point detection under high-dimensionality. The main factor of this computational complexity is the sum over all combinations of the latent classes $\mathbf{z}_{1:t-1}$. To avoid this marginalization while keeping the latent variable methodology, we simplify the previous model by considering an alternative strategy. If collapsing the entire latent domain is too costly, we propose to *observe* it [23]. That is, we can explicitly observe point estimates of z_t, which were hidden before. Concretely, instead of marginalizing $\mathbf{z}_{1:t}$, we directly plug in the values taken by each z_t.

To adapt the hierarchical model to these new *observed* latent variables, we take the set of point estimates $\mathbf{z}_{1:t}^\star$ as the input to the detector. Note that we further assume the likelihood to be $p(z_t^\star|\boldsymbol{\theta})$, that is, we are directly modeling the changes on the sequence of point-estimates. In particular, the

point estimates are given by the maximum-a-posteriori criterion. That is, $z_t^\star = \arg\max_{z_t} p(z_t|\mathbf{x}_t)$, for which we need to previously derive the posterior distribution $p(z_t|\mathbf{x}_t)$. In the remaining lines of this section, we assume that it is given, and show, later in Section 3, how to obtain it. The graphical model of this *point-estimate observation* (PEO) approximation is shown in Figure 1.

To perform CPD on this simplified model, we choose the likelihood function $p(z_t^\star|\boldsymbol{\theta})$ to be a categorical distribution with natural parameter $\boldsymbol{\pi}$. We place a Dirichlet prior on $\boldsymbol{\pi}$ with a single hyperparameter $\boldsymbol{\gamma}$. Then, the generative model is $z_t^\star \sim \mathrm{Cat}(\boldsymbol{\pi})$ and $\boldsymbol{\pi} \sim \mathrm{Dir}(\boldsymbol{\gamma})$,where $\boldsymbol{\pi} \in \mathcal{S}^K$ and $\boldsymbol{\gamma} \in \mathbb{R}_+^K$, with $\mathcal{S}^K$ and $\mathbb{R}_+^K$ being the K-dimensional simplex and positive orthant, respectively. Interestingly, this choice for the priors allows us to compute the predictive probabilities $\pi_t^{(r)}$ in closed form, which are given by

$$\pi_t^{(r)} = p(z_t^\star|r_{t-1}, \mathbf{z}_{1:t-1}^{\star(r)}) = \frac{\gamma_k^{(r)}}{\sum_{k'=1}^K \gamma_{k'}^{(r)}}, \ \forall r \in \{1, \ldots, t\}, \tag{12}$$

where $\gamma_k^{(r)}$ is the kth component of the parameters of the Dirichlet prior computed for the run length r_t. This closed-form expression, which is a direct consequence of the Dirichlet-Categorical conjugacy, provides a significant reduction in the computational complexity and results in a very simple method. Finally, the parameters are updated as $\gamma_k^{(r)} \leftarrow \gamma_k^{(r)} + \mathbb{I}[z_t^\star = k]$, with $\mathbb{I}[\cdot]$ being the indicator function. The PEO model is effective in terms of computational cost, while at the same time it provides a good performance, as Figure 2 shows.

The PEO model also allows us to consider missing temporal data. We consider that all missing observations follow the completely at random (MCAR) assumption, which only denotes the lack of correlation among them, and

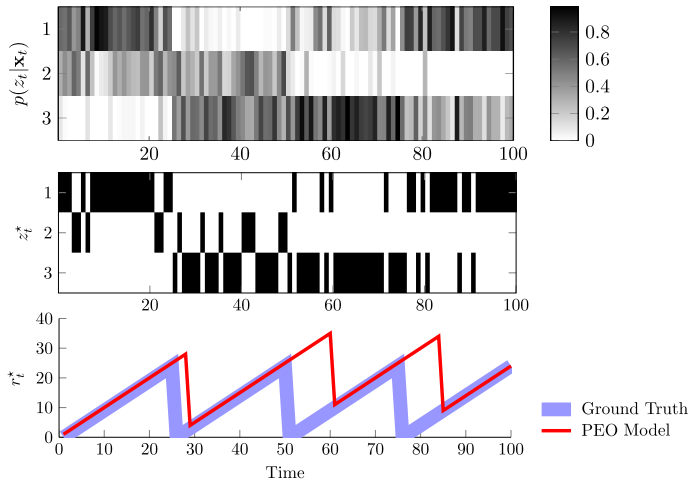

Figure 2: Performance of the PEO simplified model. **Top Row:** Sequence of posterior probability vectors $p(z_t|\mathbf{x}_t)$ where $K = 3$ and $T = 100$. Three change points are placed at $t = \{25, 50, 75\}$. **Middle Row:** Sequence of point estimates $z_t^\star = \arg\max_z p(z_t|\mathbf{x}_t)$ as 1-of-K encoding. **Bottom Row:** Ground truth of change points (blue), CP estimation $r_t^\star$ from PEO model (red).

denote them by $\mathbf{x}_t^m$. Moreover, samples that are fully observed are denoted by $\mathbf{x}_t^o$. Dividing the observations into two sets also translates to the sequence of latent variables, which are divided into lost observations z_t^m, where all components of $\mathbf{x}_t^m$ are missing, and observed variables z_t^o, which correspond to $\mathbf{x}_t^o$. The case where only some components of $\mathbf{x}_t$ are missing is addressed in Section 3 from a different perspective.

Following a Bayesian approach, we marginalize the missing latent variables out in the hierarchical model. The corresponding predictive probability is therefore reduced to $\pi_t^{(r)} = \int p(z_t^m|r_{t-1}, \mathbf{z}_{1:t-1}) dz_t^m = 1$, which still allows us to compute $p(r_t, |\mathbf{X}_{1:t-1})$, even if a certain z_t is missing. Moreover, the recursivity remains unaltered since $p(r_t|r_{t-1})$ is always evaluated sequentially.

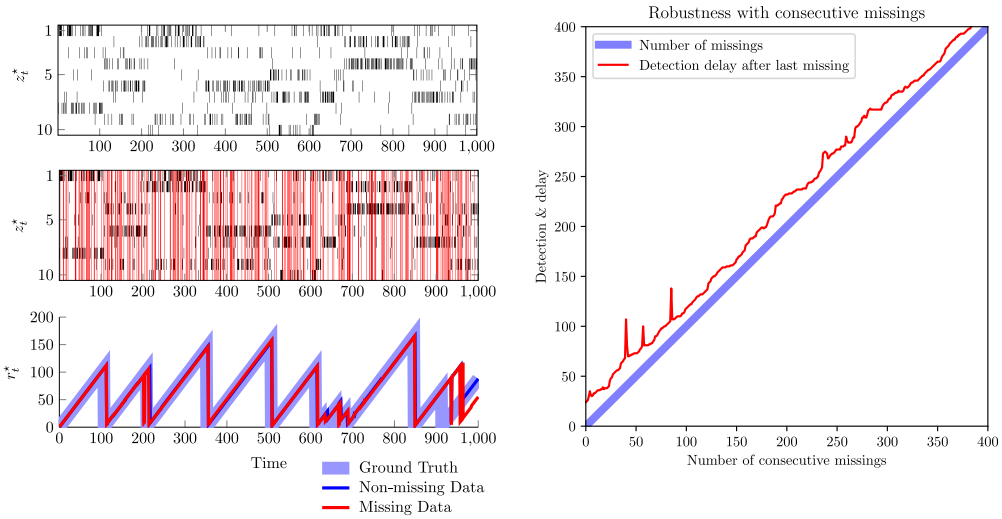

Figure 3: Change-point detection on the simplified hierarchical model with missing data. **Left Top Row:** Fully observed sequence $\mathbf{z}^{\star}_{1:t}$ of point-estimates (1-of-K encoding) with $K = 10$. **Left Middle Row:** Sequence $\mathbf{z}^{\star}_{1:t}$ with random missing entries (red) (25% rate). **Left Bottom Row:** True $r_{1:t}$ (grayed blue), MAP estimation of change points from the complete sequence (blue) and from the incomplete sequence (red). **Right:** Evolution of change-point detection delay as the number of consecutive missings increases.

That is, the uncertainty is still propagated as follows

$$p(r_t, \mathbf{z}_{1:t}) = \sum_{r_{t-1}} p(r_{t-1}, \mathbf{z}_{1:t-1}) p(r_t | r_{t-1}). \tag{13}$$

Our approach for incomplete sequences presents good performance if the missing entries do not appear as long chunks. If that were the case, the time-step t for detection would only be delayed, as the influence of very old observed data in $p(r_t, \mathbf{Z}_{1:t})$ in (13) decreases with missings.

Figure 3 compares the PEO model for complete and incomplete data sequences. First, this figure shows the MAP estimates of $\mathbf{z}_{1:t}$ for the complete case (left top row) and the incomplete one (left middle row). Additionally, in

the left bottom row we plot the estimates of r_t for missing and non-missing data, as well as the ground truth and shows that the performance with missing data is almost identical to the case of non-missing data. We also performed an analysis of the CPD robustness to periods of consecutive missing data. The results are included in Figure 3 (right). We can confirm that the method is robust under the presence of consecutive missing observations, with a similar delay as in the fully observed case. If more missing data appears, the delay augments linearly as the number of consecutive missings increases.

3. Heterogeneous Circadian Models

We now study how to embed heterogeneous models into the hierarchical BOCPD technique presented before, as well as to handle periodic temporal structures. Motivated by the application to human behavior characterization, we consider data that possess a periodic temporal structure, with 24-hours period, which is induced by the circadian rhythm. However, the ideas presented here are valid for any dataset with known periodicity.

To account for the periodic dependencies, we propose to arrange the data such that a single sample $\mathbf{x}_t$ at time t stacks the observations of one period, as shown in Figure 4. Moreover, we build heterogeneous observations by simply stacking dimensions from different statistical types on larger vectors $\mathbf{x}_t = [\mathbf{x}_t^1, \cdots, \mathbf{x}_t^M]^\top$, with M being the number of heterogeneous data types. This means that the sequence of observations $\mathbf{X}_{1:t}$ summarizes consecutive periods of different heterogeneous measurements.

For the description of one day, $\mathbf{x}_t$, we propose a heterogeneous mixture

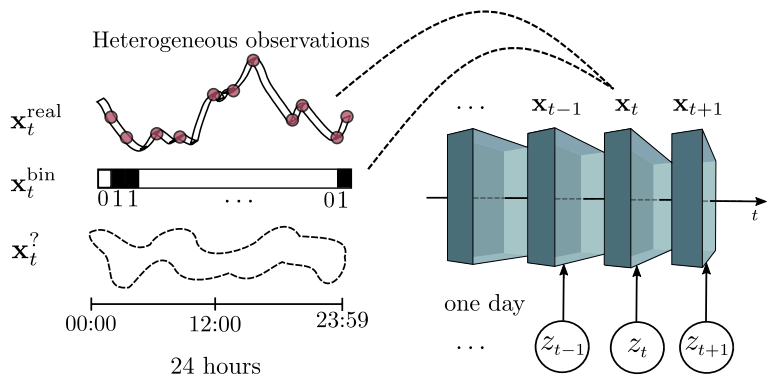

Figure 4: Schematic representation of the embedded circadian model. Each shaded box represents a daily observation $\mathbf{x}_t$. Each observation is heterogeneous, i.e., mix of real, binary values, etc. Moreover, associated to $\mathbf{x}_t$, there exists a single latent variable z_t that is lower dimensional and we aim to discover.

model for the likelihood of the form

$$p(\mathbf{x}_t|z_t, \{\boldsymbol{\phi}_k^1, \ldots, \boldsymbol{\phi}_k^M\}_{k=1}^K) = \prod_{k=1}^K \prod_{j=1}^M p(\mathbf{x}_t^j|\boldsymbol{\phi}_k^j)^{\mathbb{I}\{z_t=k\}}, \tag{14}$$

where the latent variables z_t are indicators of which component is active. Concretely, it is composed by K components, each one with its own likelihood function $p(\mathbf{x}_t^j|\boldsymbol{\phi}_k^j)$ for each data type, and with $\boldsymbol{\phi}_k^j$ denoting the parameters of the jth data type for the kth class. As can be seen in (14), we have assumed that given z_t and the parameters, the different types of observations, $\mathbf{x}_t^1, \ldots, \mathbf{x}_t^M$, are conditionally independent. This assumption simplifies the inference procedure, while at the same time it seems reasonable for the proposed application.

3.1. Circadian Gaussian-Bernoulli Mixture Model

Among all heterogeneous statistical data types that may be incorporated to the daily representation $\mathbf{x}_t$, we choose here two representative cases for

the considered application.[3] Then, we assume that each observation $\mathbf{x}_t$ is composed by binary and real values, that is, $\mathbf{x}_t = [(\mathbf{x}_t^{\text{real}})^\top, (\mathbf{x}_t^{\text{bin}})^\top]^\top$ where $\mathbf{x}_t^{\text{real}} \in \mathbb{R}^D$ and $\mathbf{x}_t^{\text{bin}} = [x_{t1}^{\text{bin}}, \ldots, x_{tD}^{\text{bin}}]^\top$, with $x_{tj}^{\text{bin}} \in \{0, 1\}$. Here, D denotes the dimensionality of each kind of observation, and is given by a function of the period. For instance, if each component of $\mathbf{x}_t^{\text{real}}$ represents the traveled distance during a given hour, then $D = 24$, which corresponds to one day, the period induced by the circadian rhythm. Note also that we have assumed that all data types have the same dimensionality, but it is straightforward to consider different dimensionalities. Next, we select Bernoulli and multivariate Gaussian likelihoods, yielding

$$\begin{aligned} p(\mathbf{x}_t | z_t = k, \{\boldsymbol{\phi}_k\}_{k=1}^K) &= p(\mathbf{x}_t^{\text{bin}}, \mathbf{x}_t^{\text{real}} | \boldsymbol{\phi}_k) \\ &= \mathcal{N}(\mathbf{x}_t^{\text{real}} | \mathbf{0}, \mathbf{K}_k + \mathbf{D}) \prod_{j=1}^{D} \text{Ber}(x_{tj}^{\text{bin}} | \mu_{kj}), \quad (15) \end{aligned}$$

where, for the time being, we denote $\boldsymbol{\phi}_k = \{\boldsymbol{\mu}_k, \boldsymbol{\gamma}_k\}$ as the set of likelihood parameters for each latent class value k. Here, $\boldsymbol{\mu}_k = [\mu_{k1}, \ldots, \mu_{kD}]^\top$, where each $\mu_{kj} \in [0, 1]$, is the mean of the jth Bernoulli variable and $\boldsymbol{\gamma}_k$ are the parameters of the covariance matrix $\mathbf{D}$, which is a common diagonal matrix for all latent classes given by $\mathbf{D} = \text{diag}(\sigma_1^2, \ldots, \sigma_D^2)$, and the parameters of $\mathbf{K}_k$. The covariance matrices $\mathbf{K}_k$ are positive-definite and correspond to each class k. The purpose of the diagonal matrix $\mathbf{D}$ is to introduce heteroscedasticity in the model since we are assuming that the noise may vary within time inputs.

To capture the periodic (circadian) feature of $\mathbf{x}_t^{\text{real}}$, we employ a periodic covariance function, similar to the works in [9] and [28]. The covariance

[3] We remark that other heterogeneous combinations can be included in the model.

matrix $\mathbf{K}_k$ is therefore generated by a non-stationary periodic kernel, i.e., $[\mathbf{K}_k]_{t,t'} = g_k(t, t')$, where the time t is assumed to be equally spaced (e.g., $t = 1, 2, 3, \ldots, D$). Further details about this function are provided in Section 3.2. Moreover, to obtain interesting insights, we consider that the distribution $\mathcal{N}(\mathbf{x}_t^{\text{real}}|\mathbf{0}, \mathbf{K}_k + \mathbf{D})$ is generated using a hierarchy similarly to [16, 17]. Concretely, the mean $\mathbf{f}$ is drawn from a zero-mean Gaussian, $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K}_k)$, which yields $\mathbf{x}_t^{\text{real}}|\mathbf{f} \sim \mathcal{N}(\mathbf{f}, \mathbf{D})$.

Following the standard approach used in mixture models, we need to compute the complete likelihood, which includes the prior probability for each latent class indicator. By defining the prior distribution as $p(z_t = k) = \pi_k$, the complete likelihood at time t is

$$p(\mathbf{x}_t, z_t = k|\{\boldsymbol{\phi}_k\}_{k=1}^K) = p(z_t = k)p(\mathbf{x}_t|z_t = k, \boldsymbol{\phi}_k) = \pi_k p(\mathbf{x}_t|\boldsymbol{\phi}_k).$$

Next, taking into account that observations are conditionally i. i. d., the complete joint log-likelihood becomes

$$\begin{aligned}\mathcal{L}_{\mathbf{z},\boldsymbol{\phi}} = \log p(\mathbf{X}_{1:t}, \mathbf{z}_{1:t}|\{\boldsymbol{\phi}_k, \pi_k\}_{k=1}^K) = \sum_{t=1}^{T}\sum_{k=1}^{K} \mathbb{I}\{z_t = k\} \log \pi_k \\ + \sum_{t=1}^{T}\sum_{k=1}^{K} \mathbb{I}\{z_t = k\} \log p(\mathbf{x}_t^{\text{bin}}|\boldsymbol{\mu}_k) + \sum_{t=1}^{T}\sum_{k=1}^{K} \mathbb{I}\{z_t = k\} \log p(\mathbf{x}_t^{\text{real}}|\mathbf{K}_k, \mathbf{D}).\end{aligned} \tag{16}$$

One final comment is in order. If we needed to take observations at non-uniformly spaced inputs $\mathbf{t}$, this approach would accept a different $\mathbf{t}$ for each $\mathbf{x}_t$ in a straightforward manner.

3.2. Periodic Non-stationary Covariance Functions

The proposed model captures the circadian feature of data for each class through the temporal embedding and the periodic covariance functions $g_k(t, t')$, which must be non-stationary in our case. Notice that afternoon-evening should have a different correlation pattern than, for instance, nocturnal hours and early morning, which prevents the use of stationary approximations. We propose to build non-stationary kernels with an input-dependent mapping $s_k(t)$, similarly to the one used in [15]. Here, we have $g_k(t, t') = s_k(t)s_k(t')\tilde{g}_k(t - t')$, where $\tilde{g}_k(t - t')$ is a stationary periodic covariance function, associated to class k, that models the intrinsic temporal structure during a day. In this case, we take the periodic version of the exponential kernel, which is given by

$$\tilde{g}_k(t - t') = \sigma_{ak}^2 \exp\Big(- \frac{2\sin^2(\pi(|t - t'|/D)}{\ell_k^2}\Big), \tag{17}$$

and for $s_k(t)$, the hour-specific term, we use a squared Fourier series of order C, with $C \leq D$. This last constraint imposes a limit on the smoothness and avoids overfitting. Thus, $s_k(t)$ is

$$s_k(t) = \left(\frac{a_{k,0}}{2} + \sum_{c=1}^{C}\left[a_{k,c}\cos\left(\frac{2\pi c}{D}t\right) + b_{k,c}\sin\left(\frac{2\pi c}{D}t\right)\right]\right)^2, \tag{18}$$

where $\mathbf{a}_k = [a_{k,0} \ \ldots \ a_{k,C}]^\top$ and $\mathbf{b}_k = [b_{k,1} \ \ldots \ b_{k,C}]^\top$ are the Fourier coefficients that parametrize the covariance matrix of the kth class, $\mathbf{K}_k$, together with the parameters of the exponential kernel, σ_{ak} and ℓ_k.

3.3. EM Algorithm for Heterogeneous Circadian Models with Missing Data

This section presents an inference technique for the model described above. In particular, given the model and the complete log-likelihood $\mathcal{L}_{\mathbf{z},\boldsymbol{\phi}}$ in (16), we

need to infer the assignments z_t and the set of model parameters $\{\pi_k, \boldsymbol{\phi}_k\}_{k=1}^K$, where we now define $\boldsymbol{\phi}_k = [\boldsymbol{\mu}_k^\top, \mathbf{a}_k^\top, \mathbf{b}_k^\top, \sigma_{ak}, \ell_k, \boldsymbol{\sigma}^\top]^\top$, with $\boldsymbol{\sigma} = [\sigma_1, \ldots, \sigma_D]^\top$.

Obtaining maximum likelihood estimates of the Gaussian-Bernoulli mixture-model parameters can be done through the expectation-maximization (EM) algorithm [8]. However, there is a slight difference in our model. While the expectation step is computed from the complete mixture distribution, in the maximization step, it is possible to factorize $\mathcal{L}_{\mathbf{z},\boldsymbol{\phi}}$ and therefore to estimate the parameters of the Bernoulli and Gaussian distributions separately. This process is equivalent for any other combination of statistical data-types.

An important aspect of using the EM algorithm here is that it allows us to handle missing components in any observation $\mathbf{x}_t$. That is, rather than dealing with sequences where we may find complete missing observations, sometimes we have to deal with just a few lost features, i.e., only a few components of $\mathbf{x}_t$ are missing. The proposed technique estimates the probabilities $p(z_t|\mathbf{x}_t)$ even if $\mathbf{x}_t$ is only partially observed. Hence, we must adapt the mixture model to deal with missing data following a similar approach to [14], which results in an algorithm that iterates between the following steps

$$\begin{aligned}
\textbf{E-step:}\ &\text{compute} \quad \mathcal{Q} = \mathbb{E}_{\mathbf{z}_{1:t}, \mathbf{X}_{1:t}^m}[\mathcal{L}_{\mathbf{z},\boldsymbol{\phi}}] \\
\textbf{M-step:}\ &\text{maximize } \mathcal{Q} \quad \text{w.r.t.} \quad \{\mathbf{a}_k, \mathbf{b}_k, \sigma_{ak}, \ell_k\}_{k=1}^K, \text{and } \boldsymbol{\sigma} \\
&\text{maximize } \mathcal{Q} \quad \text{w.r.t.} \quad \{\boldsymbol{\mu}_k, \boldsymbol{\pi}_k\}_{k=1}^K
\end{aligned}$$

where $\mathbf{X}_{1:t}^m$ denotes the matrix that collects the missing values of every observation. Finally, contrary to standard EM techniques, the proposed approach must combine closed-form estimators for the Bernoulli distribution parameters and numerical optimization for the hyperparameters of the periodic

covariance function. Since the latter problem is highly non-convex, several initializations of the hyperparameters are required achieve a good (local) maximum. If we do not allow the optimizer to converge, it would lead to the generalized version of the EM algorithm instead. More additional details about the inference process are provided in Appendices B, C, and D included in the supplementary material.

3.4. Computational Cost

The proposed hierarchical model is computationally scalable and separates the cost of having high-dimensional data from the CP detection process. The hierarchical circadian section is stable with the dimensionality D of the observations and the number of samples T. Moreover, we can scale them up without any significant cost in the computation. However, if we rise K, the size and number of parameter sets $\{\boldsymbol{\phi}_k\}_{k=1}^{K}$ may increase exponentially, depending of the likelihood model considered. The key point is that the dimension K of the discrete low-dimensional manifold must only be increased if a significant expressiveness is needed in the model. Having a fixed number of dimensions K, the computational cost of CP detection is similar to BOCPD, but independent of the likelihood model considered. That is, if the number of dimensions in the raw observations $\mathbf{X}_{1:t}$ increases, the computational cost of [1] will also increase exponentially, i.e., full-covariance Gaussian model. In our method, the computational complexity of the CP detector remains stable with only K number of sufficient statistics to be updated at each time-step t with a cost of $\mathcal{O}(K)$.

4. Experiments

In this section, we evaluate the performance of the proposed method with synthetic and real data, and also compare it with that of state-of-the-art detectors.

4.1. Synthetic data

In the first experiment, we validate the hierarchical CPD using synthetic data. We generate a discrete sequence $\mathbf{z}_{1:t}$ with $T = 500$ instances and $K = 5$ classes (future z_t assignments) where four change points were introduced. Then, the generative parameters $\boldsymbol{\pi}$ for each partition are generated by sampling $\boldsymbol{\pi} \sim \mathrm{Dir}\,(\alpha/5, \ldots, \alpha/5)$, and we fixed $\alpha = 25$. The resulting sequence $\mathbf{z}_{1:t}$ is used to generate pairs of multivariate Bernoulli-Gaussian samples following the heterogeneous model described in Section 3.

In this experiment, we only observe the given stream of binary and real observations, never the sequence of latent classes $\mathbf{z}_{1:t}$. Then, we infer the posterior probabilities over class assignments $p(z_t|\mathbf{x}_t^{\mathrm{bin}}, \mathbf{x}_t^{\mathrm{real}})$ by applying the EM algorithm described in the previous section. In the inference, we have truncated the number of Fourier coefficients to $C = 3$, while the data was generated with $C = 2$. From the sequence of posterior probability vectors $p(\mathbf{z}_{1:t}|\mathbf{X}_{1:t})$, the point-estimates $\mathbf{z}_{1:t}^{\star}$ were obtained using the MAP estimator. We run the whole process for several K classes in the Gaussian-Bernoulli mixture model and multiple initializations.

Figure 5 shows $z_t^{\star}$ and $r_t^{\star}$ for $K = 3, 4$, and 5, as well as the ground truth. The proposed method detects with high accuracy the CPs, although, as the number of categories decreases, the detection presents a larger delay. Our

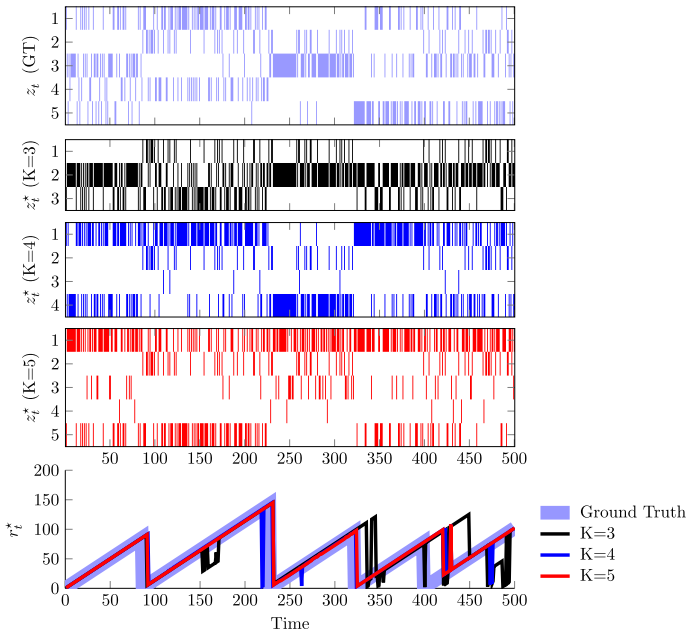

Figure 5: Results for the synthetic experiment. **First Row:** Each observation $\mathbf{x}_t$ consists of $\mathbf{x}_t^{\text{real}} \in \mathbb{R}^D$ and $\mathbf{x}_t^{\text{bin}} \in \{0,1\}^D$, with $D = 24$. **Second Row:** Sequence of true latent variables. **Third to Fifth Rows:** Sequence of MAP estimates of the latent variables for $K = 3, 4$, and 5. **Last Row:** Ground Truth of CP segments and MAP-CPD traces of $r_t^\star$ for $K = 3, 4$, and 5 latent classes. Extra Figures of simulations are included in the Appendix E within the supplementary document.

hypothesis is that as long as the circadian model gets a sufficiently large number of latent classes (i.e., it is more representative), the CP detector better identifies adjacent partitions. Additionally, it is possible to choose the model order in the hierarchical-circadian model, K, using the Bayesian and Akaike Information Criterion (BIC, AIC). Table 1 shows the average and standard deviation results obtained with ten different initializations per order. Additional figures with more simulations results are included in Appendix E of the supplementary material.

order	$\mathcal{Q}$/Log-Lik. (av. $\pm$ std.)	BIC	AIC
$K=2$	$-20.87\times10^3\pm\ 23.83$	-21.11×10^3	-20.97×10^3
$K=3$	$-20.36\times10^3\pm158.45$	-21.11×10^3	-20.92×10^3
$K=4$	$-20.35\times10^3\pm146.30$	-21.10×10^3	-20.88×10^3
$K=5$	$-20.26\times10^3\pm\ 56.41$	$\underline{-20.77\times10^3}$	$\underline{-20.46\times10^3}$
$K=6$	$-20.27\times10^3\pm\ 27.22$	-20.86×10^3	-20.49×10^3
$K=7$	$-20.24\times10^3\pm\ 62.18$	-20.99×10^3	-20.56×10^3

Table 1: Average log-likelihood, and largest BIC and AIC metrics for 10 initializations. Best BIC and AIC models are <u>underlined</u>.

4.2. Hierarchical CPD Evaluation and Comparison

This section evaluates and compares the performance of the proposed method with other CPD techniques in the literature: i) Bayesian online change-point detection (BOCPD) [1], ii) principal component analysis (PCA) preprocessing followed by BOCPD (PCA-CPD), iii) optimal partitioning (OP) and iv) binary segmentation (BS).[4] Since the BOCPD and the PCA-CPD methods work directly on the observations $\mathbf{X}_{1:t}$, in the comparison with the proposed detector, we have considered only Gaussian observations. On the contrary, the OP and BS methods work on $\mathbf{z}^{\star}_{1:t}$, and we can therefore use the Gaussian-Bernoulli model. Moreover, for the BOCPD method we have assumed a diagonal, but otherwise arbitrary covariance model, since a full-covariance model would present poor detection performance as the method cannot obtain enough evidence. Precisely this problem is what we try to

[4]References to OP and BS methods are provided in the supplementary material.

avoid by combining the BOCPD method with a dimensionality-reduction step based on PCA. We keep the two dimensions with largest variance, which yields a transformed observation with only two components and then, we use a diagonal covariance model within the BOCPD algorithm. The spirit of this dimensionality-reduction technique is similar to that in this work, but it considers a continuous low-dimensional manifold and an orthogonal linear transformation.

We evaluate the performance of each method in Table 2 for Gaussian observations of dimension D ($\mathcal{N}_D$) and the Gaussian-Bernoulli model, where each kind of observation is also of dimension D ($\mathcal{NB}_D$). Then, in the latter case, the observations have a total dimension of $2D$. Moreover, we have considered two metrics, delay and CP detection rate, for which we consider that a CP is detected if the method identifies the CP at the time it appears or afterwards. The data in all experiments were synthetically generated as in the previous section and we set five CPs every $t = 100$ steps.

The results in Table 2 show that the proposed method present higher detection rate than the competitors with sufficiently small delay. Additionally, its main strength compared to the original model of [1] is that it allows to consider extremely high-dimensional observations, heterogeneous features with different data-types and even circadian phenotypes without decreasing the detection perfomance.

4.3. Anomalous Human Behavior

In this experiment, we evaluate the proposed method on real data for medical applications. Since our work is highly motivated by the problem of change-point detection for personalized medicine, it is important to motivate

	BOCPD		PCA-CPD		HIERCPD	
model	delay	rate	delay	rate	delay	rate
$\mathcal{N}_2$	$\underline{22.30 \pm 20.02}$	$\underline{0.92}$	24.35 ± 25.51	0.68	22.31 ± 18.79	0.88
$\mathcal{N}_5$	18.41 ± 9.88	0.96	20.42 ± 13.57	0.76	$\underline{12.56 \pm 6.51}$	$\underline{1.0}$
$\mathcal{N}_{10}$	10.54 ± 8.85	0.96	14.45 ± 12.21	0.8	$\underline{8.71 \pm 4.83}$	$\underline{0.96}$
$\mathcal{N}_{20}$	$\underline{7.42 \pm 7.61}$	0.84	15.45 ± 10.22	0.88	8.76 ± 5.35	$\underline{1.0}$
$\mathcal{N}_{50}$	$\underline{5.05 \pm 2.97}$	0.76	11.81 ± 10.49	0.8	8.39 ± 3.34	$\underline{1.0}$
$\mathcal{N}_{100}$	$\underline{6.36 \pm 1.93}$	0.76	13.75 ± 13.47	0.8	9.56 ± 7.10	$\underline{1.0}$
$\mathcal{N}_{200}$	$\underline{4.41 \pm 1.18}$	0.6	12.06 ± 9.45	0.64	13.39 ± 12.27	$\underline{1.0}$

	OP		BS		HIERCPD	
model	delay	rate	delay	rate	delay	rate
$\mathcal{NB}_2$	4.26 ± 6.06	0.6	$\underline{1.25 \pm 0.95}$	0.16	25.75 ± 21.26	$\underline{0.8}$
$\mathcal{NB}_5$	2.07 ± 2.32	0.56	$\underline{1.02 \pm 1.22}$	0.2	15.19 ± 9.98	$\underline{0.84}$
$\mathcal{NB}_{10}$	$\underline{0.83 \pm 1.21}$	0.72	2.01 ± 1.82	0.16	13.86 ± 7.32	$\underline{0.92}$
$\mathcal{NB}_{20}$	$\underline{1.06 \pm 1.38}$	0.64	2.50 ± 1.73	0.16	13.08 ± 10.35	$\underline{1.00}$
$\mathcal{NB}_{50}$	$\underline{1.01 \pm 0.97}$	0.68	1.01 ± 1.58	0.36	11.21 ± 6.67	$\underline{1.00}$
$\mathcal{NB}_{100}$	1.91 ± 3.26	0.84	$\underline{0.77 \pm 0.83}$	0.36	13.83 ± 13.58	$\underline{0.96}$
$\mathcal{NB}_{200}$	$\underline{1.61 \pm 2.06}$	0.84	3.04 ± 5.81	0.36	13.54 ± 8.91	$\underline{0.96}$

Table 2: Hierarchical CPD (HIERCPD) vs. benchmark of change-point detection methods: Bayesian change-point detection (BOCPD), principal component analysis (PCA-CPD), optimal partitioning (OP) and binary segmentation (BS). The upper table shows the results for Gaussian observations with different dimensionality ($\mathcal{N}_D$), and the lower table shows the case with heterogeneous data Gaussian-Bernoulli ($\mathcal{NB}_D$). Best delay and rate metrics are underlined.

why detecting abnormal changes is a key point for the future of psychiatry.

4.3.1. Change-Point Detection in Psychiatry

Nearly all patients with mental health disorders, such as schizophrenia or chronic depression, are likely to suffer some sort of relapse in the next five years after the first episode. However, despite the fact that significant early indicators may appear before an abrupt relapse, when recognizable symptoms are already present, it is usually too late for any preventive treatment.

The majority of these previous signs and triggering effects can be identified by continuously monitoring changes in the patient's behavior. Here, we refer to human behavior as the set of routines and patterns that are persistent during our daily life. In [10], it was demonstrated how these behavioral patterns are projected in the set of signals that we generate every day (i.e., mobility data, phone communications and logs of social interactions, etc.) and are mainly recognizable through the appearance of circadian features and periodicities. Along these lines, the ubiquitous condition of smartphones and wearables, whose recording and memory capabilities are able to perform continuous monitoring of the aforementioned domains, opened a new window for analyzing all these sources of information in chronic patients without the need of direct medical interventions. Our primary goal would be to provide a robust solution to this paradigm by applying CPD on behavioral data obtained from monitored psychiatric patients.

4.3.2. Mobility Metrics

In this experiment, we used data that consists of location traces (latitude-longitude) recorded via the smartphone of a student of our lab during 275

consecutive days. It contains a bit more than 100,000 instances that correspond to the user's GPS coordinates every 3 minutes on average. We considered two types of metrics [6]: 1) a real-valued signal of the log-distance travelled per hour and 2) daily binary vectors of presence or absence at home.

The preprocessing step was different for each one of the metrics. In the first case, we collected all location traces separately for every hour and every day. Once this was done, we calculated the approximate distance traveled between consecutive latitude-longitude pairs. It is important to remark that if the maximum time without any location point, which we set to 30 minutes, was exceeded for some specific hour, the total distance in that interval was considered as a missing value. For the binary case, we preprocessed all the locations in order to estimate the home location, that is, the most usual latitude-longitude pair during nocturnal hours. To decide whether the student was or not at home, we established a maximum distance range of 50 meters. When one or more location traces show a relative distance to the estimated home location equal or lower than this range, then we assume that at this hour the student was at home. For the hours with no location traces, we considered them also as missing data.

After the preprocessing step, we can see both signals in Figure 6 (upper plots), which correspond to the observations $\mathbf{X}_{1:t}$ in the circadian model. For the binary metric, we have 24-dimensional vectors for each day where at-home (black) and not-at-home (white) indicators are showed, in contrast to missing values (red). An important detail to remark is that missing traces are mainly concentrated at night. This may occur when smartphones run out of battery during night, which a priori could make sense.

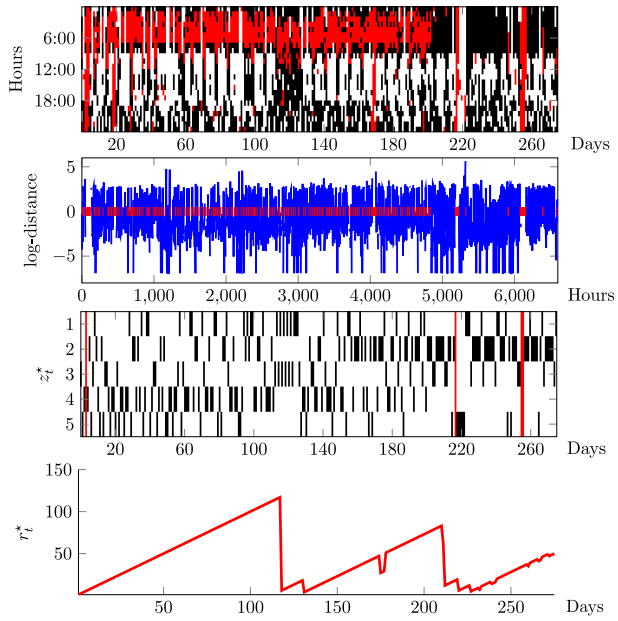

Figure 6: Results from the experiment with real data. **First and Second Rows:** Binary and real valued observations per day and hour, respectively. Red traces correspond to missing hours. **Third Row:** MAP estimates of the latent class indicators, $z_t^\star$, inferred by the PEO model. Red lines are missing latent variables, which correspond to days where all features are lost. **Fourth Row:** MAP estimates of the run length using the most representative model, i.e., $K = 5$ classes.

4.3.3. Detection of Behavioral Changes

The results of the EM algorithm after observing the sequence of heterogeneous data are shown in Figure 6. Similarly to the previous synthetic experiment, the main objective is to obtain the posterior probabilities $p(\mathbf{z}_{1:t}|\mathbf{X}_{1:t})$. Then, based on the PEO model, we set the point estimates $\mathbf{z}_{1:t}^\star$ as the sequence of observed latent variables, which are shown in the third row of Figure 6.

Results using AIC and BIC model selection indicate that $K = 5$ classes is the optimal complexity. The hierarchical model shows that the second

and fourth profiles are the most frequent. In the first partition, the natural parameters $\boldsymbol{\pi}_1$ indicate that the fourth and fifth profiles mainly dominate the routine. However, after a small secondary partition given by an unusual combination of classes, the third partition shows a new pattern of behavior where the second profile is the most frequent one. We confirm that the three main CPs detected, respectively at days $t = 117, 132$ and 210 have a ground-truth interpretation for the student under study. Particularly, the short partition between CP_1 and CP_2 (period $\rho = [117, 132]$) corresponds to *pre*-deadline days during holidays, explaining the shift in both mobility and sleep cycles (see Figure 6). We also identify CP_3 at $t = 210$ and the consecutive days, which are actually irregular, as the week of *easter* holidays, when the student was out of the laboratory. Importantly, we have found that our method is able to: 1) infer different patterns of routines as, for instance, job/leisure behaviors, 2) CPs appear due to extreme discrepancies between partitions and is robust to outliers (i.e., weekends, uncommon working days) and 3) distinguish (potentially) similar behaviors, e.g. partitions $\rho_1 = [1, 117]$ and $\rho_3 = [132, 210]$ that corresponds to working epochs but with the difference of being *Autumn* and *Spring* seasons (see patterns of $z_t^\star$ at Figure 6).

4.3.4. Circadian Phenotypes

One of the strengths of the latent model is its ability to identify the 24-hours periodic patterns from the set of mobility metrics, by estimating the Fourier coefficients of the covariance functions for the distance values.

After fitting all hyperparameters in the maximization step during the inference, we are able to reproduce the patterns that describe how a person usually behaves during each type of day. In Figure 7, we represent the set of

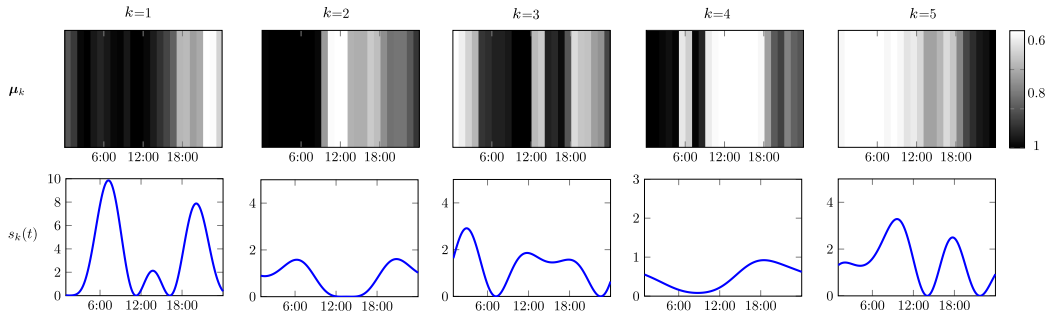

Figure 7: Circadian patterns of mobility. The results are obtained for $K = 5$. **Left Column:** Periodic functions generated from the squared Fourier series with the estimated hyperparameters. Functions represent the variance in the mobility per hour and type of day. **Right Column:** Thermal plots of multivariate Bernoulli distribution, i.e., the probability of being at-home during different hours.

(multimodal) periodic functions generated from the squared Fourier series, $s_k(t)$, and the estimated vectors of probability $\boldsymbol{\mu}_k$ of the Bernoulli likelihood.

5. Conclusions

In this paper we have introduced a novel generalization for Bayesian change-point detection methods to handle heterogeneous high-dimensional observations with unknown periodic structure. The proposed technique makes two contributions. The first one, denoted as the hierarchical detector, is a probabilistic extension of the widely known Bayesian online change-point detection (BOCPD) algorithm presented in [1]. We extended the method to accept any type of latent variable model as the underlying predictive model in the change-point detector. The main benefit of the strategy is that the detector now works with less complex discrete data, that is, the latent class representations where the change points are easier to be detected. Another

key point is that the cost of inferring such latent representations is assumed by the hierarchical model, which does not affect the detector.

Secondly, we present a heterogeneous circadian mixture model that can be embedded with the detector in order to capture all underlying periodicities in the temporal data. The main strength of this novel model is that it may accept any combination of statistical data types in the original set of observations as well as non-stationary periodic samples, whose main patterns and conditional periodicities are captured via covariance functions or kernels. In this paper, we particularize the heterogeneous approach to the case of mixing Gaussian and Bernoulli distributed variables. However, if other combinations were considered, the computational complexity of inference in the surrogate hierarchical model would increase according to the number of parameters induced by the new distributions. The main benefit is that the performance of the change-point detector would not be affected by the new arbitrary combination of statistical data-types.

Lastly, we have also introduced a new approach for detecting change points even from sequences with multiple missing values. This solution can be used in any variation of the BOCPD method, and is specially useful when using latent variable models. In addition, we have also configured the model to capture the 24 hours periodic structure of the data for applications where we must focus on the circadian rhythm for human behavior characterization.

The code is publicly available at `https://github.com/pmorenoz/HierCPD`. The package includes the heterogeneous circadian mixture model, as well as the EM inference and the periodic non-stationary covariance functions for reproducibility purposes.

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